MARCH 24TH 2026

ALY JUMA - HEAD OF DATA - CREDABLE GROUP

Credable

Hackathon Kick-off

Credable Group

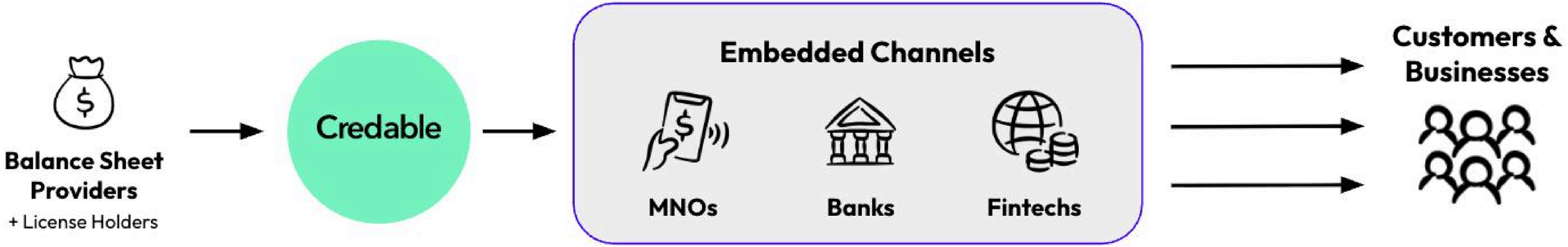
A Digital Banking Platform partnering with channels to deploy credit and savings products at scale.

HACKATHON KICK-OFF

Credable

How We Work

Our embedded partnership model, means Credable acts as the enabler for capital providers and channel partners to deploy financial products at scale.



Our Data Practice

Thin File Decisioning

- We use alternative data to build risk scores for customers traditional bureaus can't see
- Sources include mobile money, bank transactions, CRB, utility and behavioural signals
- Models for default, churn, and lifetime value feed into a single credit decision

Why This Matters

- Alternative data is one of the only paths to formal credit for the 1.4 billion unbanked
- Predicting financial distress helps us lend responsibly and protects partners from default and churn
- Customer stability is the foundation of sustainable financial inclusion

The data you're working with is a similar signal Credable uses in production — mobile money behaviour as a proxy for financial health.

The Data Set

Anonymized, synthetic monthly mobile money data on 10,000 customers.

01

General Overview

70,000 records
226 Columns
7 MM Transaction Categories
Rolling 6 Month Snapshot

02

Customer Profiles

Customer ID
Gender
Region
Age
Segment
ARPU
Smartphone
Earning Pattern
Activity Rate

03

Category Signals

Payments
Merchants
Transfer from Bank
Deposits
Withdraw
Send
Receive

The Data is not Perfect... by Design

Each monthly snapshot contains 6 months of historical behaviour — but don't assume the history is consistent across snapshots. When in doubt, **the most recent month's (M1) data is your most reliable foundation**. The data rewards those who interrogate it carefully.

In the real world, raw financial data is never analysis-ready. How you understand the structure, engineer your features, and validate your assumptions is just as important as the model you build.

Things to Know

- Each category carries 5 signals per month: volume, total value, highest single transaction, counterparty count (companies/merchants/agents/recipients), and last activity date — giving you both how much and how often for every behaviour type.
- **Nulls are meaningful** — a null across an entire category for a month isn't missing data, it means zero activity. That absence is itself a signal
- The rolling window shifts — each monthly snapshot shows the prior 6 months (M1–M6), so the same customer appears 7 times with the window advancing each time
- Daily average balance (M1–M6_DAILY_AVG_BAL) is a separate field, not part of the 7 categories — it's a key standalone feature
- **Profile fields:** ARPU, age, gender, region, smartphone ownership, segment (LVC/MVC/HVC), and earning pattern are fixed per customer across all snapshots
- 90D_ACTIVITY_RATE captures how consistently active the customer has been — a useful baseline engagement signal

Data Dictionary

Field	Description
date	Snapshot date (1st of each month, Jan–Jul 2024)
customer_number	Unique customer identifier
arpu	Average revenue per user
age	Customer age
gender	Customer gender (M/F)
region	Geographic region
smartphone	Smartphone ownership (Yes/No)

Data Dictionary (continued)

Field	Description
segment	Customer value tier (LVC / MVC / HVC)
earning_pattern	Income cadence (e.g. Weekly Earner)
90d_activity_rate	Share of days active in the last 90 days
Mx_DAILY_AVG_BAL	Average daily wallet balance in month x

Data Dictionary (continued)

Field	Description
PER CATEGORY TRX TYPES	REPEATED FOR M1–M6 ACROSS ALL 7 TRANSACTION TYPES
Mx_[TYPE]_VOLUME	Number of transactions in month x
Mx_[TYPE]_TOTAL_VALUE	Total value of transactions in month x
Mx_[TYPE]_HIGHEST_AMOUNT	Single largest transaction in month x
Mx_[TYPE]_[COUNTERPARTY]	Unique counterparties in month x (companies / merchants / banks / recipients / agents)
Mx_[TYPE]_LAST_DATE	Date of most recent transaction in month x

Transaction Types

Field	Description
PAYBILL (outflow)	Mobile money transactions to paybills (payments)
MERCHANTPAY (outflow)	Mobile money transactions to merchants
TRANSFER_FROM_BANK (inflow)	Mobile money transaction from bank
MM_SEND (outflow)	Mobile money transaction sent to another user (P2P)
RECEIVED (inflow)	Mobile money transaction received from another user (P2P)
DEPOSIT (inflow)	Mobile money cash deposit into account
WITHDRAW (outflow)	Mobile money cash withdrawn from account

THE CHALLENGE

PREDICTING FINANCIAL DISTRESS

About the Challenge

AI4EAC INNOVATION CHALLENGE

Challenge Design

AI4EAC Liquidity Stress Early Warning Challenge

The task: Build a machine learning model to predict which customers are likely to experience financial stress within the next 30 days.

Data: Tabular

Multi-Metric Evaluation:

- Log Loss (60%)
- ROC-AUC (40%)

Submission Format requirements: ID columns + 2 columns [TargetLogLoss, TargetRAUC]

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AI4EAC INNOVATION CHALLENGE

Dataset Overview

- You are given a dataset consisting of mobile money transactions over a six month period
  - Mobile money inflows and revenue
  - Cash withdrawals
  - Balance levels
  - Merchant and bill payments
  - Transaction frequency

- Dataset split
  - **Train: 40,000 rows**
  - **Test: 30,000 rows**

- Class Distribution
  - **~10,500 experiences liquidity stress**

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How Log Loss Works

Core Idea:
Measures how well predicted probabilities match actual outcomes.

How it Works

- You report a probability
- -pred_prob*log(y) - (1 - pred_prob)*log(1 - y):

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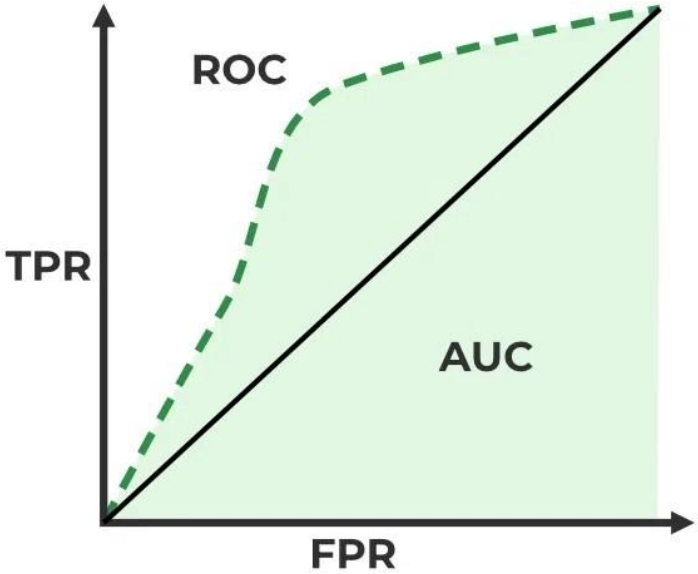
How ROC-AUC Works

Metric

- Measures how well the model ranks stressed customers above non-stressed customers for each threshold:

- ROC (Receiver Operating Characteristic) curve is used to visualise a classification model performance. The AUC (Area Under the Curve) provides a score between 0 and 1 representing the probability of correctly ranking positive examples higher than negatives ones.



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Preparing for the Challenge

Practice Challenge: [AI4EAC Finance Practice Challenge]

The task: Predict the probability of loan default

Evaluation Metric:

- **F1-Score**

Key Preparation activities:

- Familiarise yourself with Zindi
- Understand the submission process
- Deal with imbalanced datasets

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